

Invariant Quotients in SEPAR: Practical Key Recovery with Chosen IVs and Repeated Resets

Noah Ostle
University of Technology Sydney

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Abstract

We analyze the reference implementation of SEPAR, a stateful cipher with a 256-bit key and a 16-bit data path. Because four assignments in its diffusion layer are executed sequentially, the implemented map is $L(a, b, c, d) = (b \oplus c, b, a, a \oplus d)$. Every keyed stage is consequently triangular: one nibble evolves autonomously, while the high byte, middle byte, and top 12 bits form exact permutation quotients. Only $(16!)^{4129}$ of the $65536!$ permutations on a word have this form. The key schedule further creates large, non-uniform fibres, the eight stages use disjoint 32-bit key pairs, and the nominal ninth state word is determined by the other eight.

We use these properties to factor key recovery from the outside in. A fixed-IV prefix experiment orders candidates for the last stage. Thereafter the whole attack is described by one pivot, $C_t = R_t + 2^8 h_{t+1}$: invert a candidate stage, remove the appropriate state byte, and recurse. We prove a bidirectional carry bound that never rejects the true path, and finish stage one with an exact constant-difference test. Rankings affect only traversal order. If every score tier is visited and early stopping is removed, the search reaches the original key for every instance; a reachable example in which the true outer lane ranks second shows why no rank-one assumption is permissible.

The threaded C proof of concept targets a repeated-reset, chosen-IV oracle. With one score tier at each of its three ranking levels, it recovered 95 of 100 preselected random keys, conditional on the correct last-stage pair. The exact one-sided 95% lower confidence bound on success is 0.8977. Successful runs took a median of 181.1 seconds on the test host. The matching outer-stage experiment selected the correct pair first for all 100 keys, although this is an empirical observation rather than a universal claim.

Keywords: lightweight cryptography; invariant quotient; chosen IV; key recovery; SEPAR.

1 Introduction

SEPAR combines eight keyed 16-bit permutations with eight state words and an LFSR-like word [3]. Its security discussion is based on local S-box properties, active-S-box counts, a nominal 144-bit state, and statistical tests of the output. Later work multiplies local differential bounds across stages [4, 5]. These arguments do not rule out an invariant partition of a complete implemented stage.

The implementation has several such quotients. These are not low-probability trails: equality in the quotient is preserved for every input and every key. A difference-distribution table for individual word differences can therefore look satisfactory even though a proper partition remains closed. The 16-bit data path also makes a complete reset codebook cost only 2^{16} chosen messages. Finally, the master key is divided into eight disjoint 32-bit pairs, so an outer stage can be removed without guessing any inner key material. Similar small-block designs have exhibited related structural failures [2].

The main results are as follows.

1. We derive the implemented diffusion layer and prove that every keyed stage is triangular. This gives exact quotient partitions and a direct counting distinguisher.
2. We characterize the autonomous nibble, enumerate the fibres created by the lossy key derivation, and show that initialization has 128, rather than 144, reachable bits for each fixed key.
3. We give an inward key-recovery algorithm based on a single pivot and a proved bidirectional carry filter. Exhaustive traversal is complete without a ranking assumption; bounded traversal is measured statistically.

The target throughout is the distributed `SEPAR/main.c`. In particular, the nibble assignments are sequential and two subkey updates use bitwise OR. Replacing either operation would define a different cipher.

2 Cipher and adversarial model

All additions and subtractions are in $\mathbb{Z}/2^{16}\mathbb{Z}$, while $\oplus$ denotes XOR. Bits are numbered from zero at the least-significant bit. Write a word as nibbles (a, b, c, d) , most significant first. The key is $K = (K_1, \dots, K_8)$, with disjoint, independently used $K_i = (k_{i,0}, k_{i,1}) \in \mathbb{Z}/2^{16}\mathbb{Z}^2$.

Let S_1, S_2, S_3, S_4 be the four coordinate S-boxes, and Σ their parallel application. In hexadecimal, the implemented tables are

x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
S_1	1	F	B	2	0	3	5	8	6	9	C	7	D	A	E	4
S_2	6	A	F	4	E	D	9	2	1	7	C	B	0	3	5	8
S_3	C	2	6	1	0	3	5	8	7	9	B	E	A	D	F	4
S_4	D	B	2	7	0	3	5	8	6	C	F	1	A	4	9	E

Define

$$G(u) = u \text{ OR } (S_1((u \gg 6) \& 15) \ll 6). \quad (1)$$

At stage i , the source derives

$$\begin{aligned} \kappa_0 &= k_{i,0}, & \kappa_1 &= k_{i,1}, \\ \kappa_2 &= G(\text{rol}_{16}(k_{i,0}, 6)) \oplus (i + 2), & \kappa_3 &= G(\text{rol}_{16}(k_{i,1}, 10)) \oplus (i + 3). \end{aligned} \quad (2)$$

For $X_u(x) = x \oplus u$, the exact stage is

$$\begin{aligned} E_i &= X_{\kappa_2 \oplus \kappa_3} \circ \Sigma \circ X_{\kappa_0 \oplus \kappa_1} \circ L \circ \Sigma \circ X_{\kappa_3} \circ L \circ \Sigma \\ &\quad \circ X_{\kappa_2} \circ L \circ \Sigma \circ X_{\kappa_1} \circ L \circ \Sigma \circ X_{\kappa_0}. \end{aligned} \quad (3)$$

Here L is the diffusion function derived in Section 3. All components are bijective; `DEC_Block` implements $D_i = E_i^{-1}$.

For initialized state words $s_1, \dots, s_8$, define

$$\begin{aligned} R_0(x) &= x, \\ R_t(x) &= E_t(R_{t-1}(x) + s_t), \quad 1 \leq t \leq 8. \end{aligned} \quad (4)$$

The reset one-word ciphertext is $R_8(x)$. The in-place state update after a word matters for multiword transcripts. The outer experiment requires only the source identity $s'_8 = s_8 + v_{45}$, where v_{45} is the stage-four output.

The adversary may choose and reuse a 128-bit IV, reset before each chosen message, and submit chosen sequences of 16-bit words. This is a repeated-reset/chosen-IV model, not a nonce-respecting model; applicability therefore depends on an interface that permits nonce reuse or arbitrary state reset. No decryption oracle is needed. The outer experiment uses two-word messages; inward recovery uses complete reset one-word codebooks at a deterministic public IV family. Two stateful 64-word messages are reserved for verification. The proof-of-concept programs contain a local oracle; the secret is used to generate replies and, when requested, to report experimental ranks, but it does not enter a search or acceptance decision.

3 Structural weaknesses

3.1 Sequential linear layer

The source extracts (a, b, c, d) , executes sequentially

$$a \oplus = c, \quad b \oplus = d, \quad c \oplus = b, \quad d \oplus = a,$$

then XORs the packed result with rotations by 12 and 8 bits.

Lemma 1 (Implemented diffusion).

$$L(a, b, c, d) = (b \oplus c, b, a, a \oplus d), \quad (5)$$

$$L^{-1}(A, B, C, D) = (C, B, A \oplus B, C \oplus D). \quad (6)$$

The nibble branch number is two.

Proof. After the four assignments the tuple is $(a \oplus c, b \oplus d, c \oplus b \oplus d, d \oplus a \oplus c)$. Substitution into the two rotations and cancellation gives (5); substitution gives the inverse. A difference confined to d remains confined to the fourth output nibble, attaining branch number $1 + 1$. Bijectivity maps every nonzero input difference to a nonzero output difference, so no sum below two is possible. $\square$

One keyed S-box-plus-linear round has dependencies

$$\begin{aligned} a' &= S_2(b \oplus \beta) \oplus S_3(c \oplus \gamma), & b' &= S_2(b \oplus \beta), \\ c' &= S_1(a \oplus \alpha), & d' &= S_1(a \oplus \alpha) \oplus S_4(d \oplus \delta). \end{aligned} \quad (7)$$

Theorem 2 (Triangular stage). *For every stage and key there are permutations on $\mathbb{Z}/2^4\mathbb{Z}$ such that*

$$\mathbf{E}_i(a, b, c, d) = (P_b(a), Q(b), R_b(c), T_{a,b,c}(d)). \quad (8)$$

Proof. Four applications of (7) yield

	a	b	c	d
1	$\{b, c\}$	$\{b\}$	$\{a\}$	$\{a, d\}$
2	$\{a, b\}$	$\{b\}$	$\{b, c\}$	$\{a, b, c, d\}$
3	$\{b, c\}$	$\{b\}$	$\{a, b\}$	$\{a, b, c, d\}$
4	$\{a, b\}$	$\{b\}$	$\{b, c\}$	$\{a, b, c, d\}$.

Final S-boxes and XORs add no dependencies. Bijectivity of $\mathbf{E}_i$, after fixing preceding triangular coordinates, forces each one-coordinate component to be injective and hence a permutation. $\square$

Define

$$\text{hi}(x) = x \gg 8, \quad \text{mid}(x) = (x \gg 4) \& 255, \quad \text{lane}(x) = (x \gg 8) \& 15, \quad \text{top}_{12}(x) = x \gg 4.$$

Corollary 3 (Exact quotient partitions). *For each $\pi \in \{\text{hi}, \text{mid}, \text{lane}, \text{top}_{12}\}$ there is a permutation $q_{i,K,\pi}$ satisfying*

$$\pi(\mathbf{E}_i(x)) = q_{i,K,\pi}(\pi(x)) \quad \text{for every } x \in \mathbb{Z}/2^{16}\mathbb{Z}. \quad (9)$$

The same holds for $\mathbf{D}_i$. Equality in each quotient is therefore preserved with probability one in both directions.

Proof. The triangular theorem gives well-defined maps on the partitions. A quotient collision would send two equal-size input fibres into one output fibre, contradicting bijectivity. Inversion gives the statement for $\mathbf{D}_i$. $\square$

Thus, for example, an input difference confined to the low byte remains in the low byte with probability one. This does not contradict small probabilities for particular full-word differences; it shows that full-word trails are the wrong projection.

3.2 Counting the defect

There are $16!$ choices for Q , $(16!)^{16}$ each for P_b and R_b , and $(16!)^{16^3}$ for $T_{a,b,c}$.

Proposition 4 (Permutation-family bound). *Exactly*

$$|\mathcal{T}| = (16!)^{1+16+16+4096} = (16!)^{4129} \quad (10)$$

permutations of $\mathbb{Z}/2^{16}\mathbb{Z}$ have form (8). Hence a uniform random permutation lies in $\mathcal{T}$ with probability

$$\frac{(16!)^{4129}}{65536!} < 2^{-771328.03}.$$

Preserving only the top-12-bit partition already has probability below $2^{-729538.24}$.

Proof. Independently chosen component permutations give distinct bijective triangular maps and every such map reveals its components. Divide by $65536!$. For the last claim, a partition-preserving permutation has exactly $4096!(16!)^{4096}$ choices: permute the 4096 fibres and then permute the 16 points independently inside each fibre. $\square$

3.3 Autonomous lane and lossy key fibres

Let β_j be bits 11–8 of κ_j . Put

$$p_0 = S_2(b \oplus \beta_0), \quad p_1 = S_2(p_0 \oplus \beta_1), \quad p_2 = S_2(p_1 \oplus \beta_2), \quad p_3 = S_2(p_2 \oplus \beta_3).$$

The second output nibble is exactly

$$q_\beta(b) = S_2(p_3 \oplus \beta_0 \oplus \beta_1) \oplus \beta_2 \oplus \beta_3. \quad (11)$$

Stage constants affect only the low nibble. We call $\nu = (\beta_0, \beta_1, \beta_2, \beta_3)$ a *lane tuple* and write q_ν for its induced permutation.

Lemma 5 (Lane injectivity; computer-assisted). *All 2^{16} tuples β induce distinct permutations in (11).*

Proof. The supplied checker evaluates all 16 inputs for every tuple, packs each output list into 64 bits, sorts the 65,536 signatures, and compares adjacent values. This exhausts the finite domain. $\square$

The OR in (1) is non-injective. For $g(x) = x \text{ OR } S_1(x)$,

y	1	3	4	7	9	B	D	E	F
$ g^{-1}(y) $	1	1	1	2	1	1	1	3	5.

Fixing a raw word’s direct and derived lane nibbles produces 256 buckets: 128 have size 128, 64 size 192, and 64 size 576. The two raw words range over a Cartesian product.

Proposition 6 (Raw lane fibres). *Every lane tuple is realizable. Its raw 32-bit key fibre has the exact size distribution in Table 1; distinct fibres partition all 2^{32} stage keys.*

Table 1: Exact raw K_i fibre distribution.

Pairs per lane	Number of lanes
16,384	16,384
24,576	16,384
36,864	4,096
73,728	16,384
110,592	8,192
331,776	4,096

Proof. For the first raw word, one nibble fixes bits 11–8. The derived nibble fixes bits 5–4 and two bits of g applied to the low nibble; six bits remain free. The four two-bit classes contain 2, 3, 2, and 9 inputs, yielding bucket sizes 128, 192, 128, and 576. The second word has the same histogram at rotated positions. Cartesian products give the table. Every raw pair maps to one tuple, proving disjoint coverage. $\square$

4 Initialization and reachable state

For one initialization round, compute

$$\begin{aligned} v_1 &= E_1(s_1 + s_3 + s_5 + s_7), \\ v_i &= E_i(v_{i-1} + s_i), \quad 2 \leq i \leq 8, \end{aligned} \quad (12)$$

then output

$$t_1 = s_1 + v_8, \quad t_i = s_i + v_{i-1} \quad (2 \leq i \leq 8). \quad (13)$$

Theorem 7 (Explicit initialization inverse). *One round is a permutation of the eight-word state. Its inverse is*

$$\begin{aligned} s_1 &= t_1 - E_8(t_8), \\ s_i &= t_i - E_{i-1}(t_{i-1}), \quad 3 \leq i \leq 8, \\ s_2 &= t_2 - E_1(s_1 + s_3 + s_5 + s_7). \end{aligned} \quad (14)$$

Thus four rounds biject the 128-bit IV onto the eight final state words. The initialized LFSR is the deterministic value

$$\ell = E_8(t_8) \text{ OR } 0x0100. \quad (15)$$

Proof. For $2 \leq i \leq 8$, (13) gives $v_i = E_i(t_i)$. Hence $s_1 = t_1 - v_8$ and $s_i = t_i - v_{i-1}$ for $i \geq 3$. These words determine v_1 , then $s_2 = t_2 - v_1$. Substitution proves a two-sided inverse. In the last round the value assigned to the LFSR is $v_8 = E_8(t_8)$, followed by the OR. $\square$

The ninth state word adds no entropy. Exactly 2^{128} of the nominal 2^{144} encodings are reachable. This inverse is also used by the counterexample certificate below to turn a chosen reachable state into its unique IV.

5 Attack overview

The attack removes the eight keyed stages in reverse order. Its two phases use the same structural defect at different resolutions.

First, fix an IV and encrypt every possible first word followed by seven short probes. The first word runs through every possible translation before the last stage. The autonomous nibble of that stage therefore gives a small conjugacy problem: the 2^{16} lane permutations induced by subkey-nibble tuples can be scored directly from the transcript. Each such tuple identifies a disjoint fibre of raw 32-bit stage keys. Candidates within that fibre are then ordered at byte and word resolution. This produces an ordering of candidates for K_8 ; it does not assume that the true one is ranked first.

For the remaining stages, collect the complete one-word reset codebook under several fixed IVs. Regard each codebook as a table C_8 . To remove stage t , try a stage key and two state bytes, apply the inverse stage to every table entry, subtract those bytes, and obtain a candidate table C_{t-1} . A carry theorem gives four necessary conditions on this table and its inverse. Only candidates that violate one of these conditions are discarded. The same step is repeated until stage one, where a constant-difference identity tests the final key exactly. Candidate keys that reach a leaf are checked by replaying all collected codebooks and two held-out stateful transcripts.

The practical search tries only a few leading score tiers. The completeness argument is separate: visit every lane, every raw key in its fibre, and every state-byte candidate that passes the necessary conditions. Since scores are then used solely to choose an order, they cannot affect coverage. This distinction is important: Section 6 gives a reachable instance for which the true last-stage lane is not the highest-scoring one.

6 Recovering the last stage

Fix one IV. For every prefix $p \in \mathbb{Z}/2^{16}\mathbb{Z}$, query the second ciphertext word $C_p(q)$ of (p, q) for

$$q \in \{0, 1, 2, 4, 8, 15, 16\}. \quad (16)$$

During encryption of the first word, put $v_{12} = E_1(p + s_1)$ and $v_{r,r+1} = E_r(v_{r-1,r} + s_r)$ for $2 \leq r \leq 7$. The map $p \mapsto v_{45}$ is a composition of translations and permutations, hence a bijection. The source then updates $s'_8 = s_8 + v_{45}$. Exhausting p therefore visits every possible outer translation once, although the other state words remain correlated.

For the second word q , let $v_{78}(p, q)$ denote its stage-seven output from the state reached after prefix p , and put $z(p, q) = \text{lane}(v_{78}(p, q) + s'_8(p))$. If ν is the true outer lane tuple, the observed nibble is $y(p, q) = \text{lane}(C_p(q)) = q_\nu(z(p, q))$. For $\Delta = \{1, 2, 4, 8, 15, 16\}$, define

$$\begin{aligned} M[a, b] &= \#\{(p, d) : y(p, 0) = a, y(p, d) = b, d \in \Delta\}, \\ H[a, b] &= \#\{(p, d) : z(p, 0) = a, z(p, d) = b, d \in \Delta\}. \end{aligned} \quad (17)$$

Then exactly

$$M[q_\nu(a), q_\nu(b)] = H[a, b]. \quad (18)$$

The score of candidate lane tuple μ is

$$\text{sc}_4(\mu) = \sum_{j=0}^{15} M[q_\mu(j), q_\mu(j+1 \bmod 16)]. \quad (19)$$

Writing $\sigma = q_\nu^{-1} \circ q_\mu$ reduces it to

$$\text{sc}_4(\mu) = \sum_j H[\sigma(j), \sigma(j+1)]. \quad (20)$$

This identity explains the statistic, but it does not make the identity cycle maximal in H .

The same transcript is also counted at byte and word resolution. Let $M_8[u, v]$ and $M_{16}[u, v]$ count the observed high-byte and full-word edges, respectively. For a raw pair K , put $q_K(u) = \text{hi}(\text{E}_{8,K}(2^8 u))$; the quotient theorem makes the low byte of this representative irrelevant. The two lifting scores are

$$\text{sc}_8(K) = \sum_{u=0}^{255} M_8[q_K(u), q_K(u+1)], \quad (21)$$

$$\text{sc}_{16}(K) = \sum_{x \in \mathbb{Z}/2^{16}\mathbb{Z}} M_{16}[\text{E}_{8,K}(x), \text{E}_{8,K}(x+1)], \quad (22)$$

where indices wrap around their domains. These scores refine the order inside a lane fibre; they do not remove a raw key from exhaustive traversal.

6.1 Why the best-scoring lane need not be correct

The following certificate was found by search and then verified deterministically. Inverting four initialization rounds reconstructs the IV; direct computation verifies every entry of (18) and enumerates all 2^{16} lanes. The key and IV are

$K = \text{E327EAACDFC5765633A8CECCBC7D7E0}$
 $\text{CD7A3ACFFA675B06E4E7004901112801},$
 $\text{IV} = \text{8D488D31DABD103BC78AB4351BCA28F7}.$

The winning tuple 6789 scores 27,899, whereas the true tuple 6781 scores 27,747 and has rank two. Thus universal rank-one or maximality claims are false, even on a reachable initialized state.

6.2 Complete enumeration of K_8

We therefore sort all 2^{16} lane tuples by (19) with a deterministic tie order. For rank $r = 1, \dots, 2^{16}$, enumerate the complete preimage fibre of the lane-to-key map and emit every raw pair in that disjoint fibre. The high-byte score orders the fibre; the full-word score refines only its maximal high-byte tier. Every unrefined and non-maximal remainder is retained for later traversal. Proposition 6 then proves coverage of all 2^{32} outer pairs. The rank-two certificate above is also a direct regression: the first lane omits the true pair, while the second emits $K_8 = (0111, 2801)$.

7 Peeling the remaining stages

Use c deterministic reset contexts indexed by j , all sharing the key but having state words $s_t^{(j)}$. Write

$$s_t^{(j)} = b_t^{(j)} + 2^8 h_t^{(j)}, \quad 0 \leq b_t^{(j)}, h_t^{(j)} < 256,$$

and let $R_t^{(j)}$ be (4) in that context.

Definition 8 (Pivot). At depth t , the search table is

$$C_t^{(j)}(x) = R_t^{(j)}(x) + 2^8 h_{t+1}^{(j)}, \quad h_9^{(j)} = 0. \quad (23)$$

The root $C_8^{(j)} = R_8^{(j)}$ is the observed reset codebook.

Given a candidate pair K and high byte h , compute

$$W_{t-1}^{(j)}(x) = \text{D}_{t,K}(C_t^{(j)}(x) - 2^8 h), \quad (24)$$

then try a low byte b and recurse on

$$\tilde{C}_{t-1}^{(j)}(x) = W_{t-1}^{(j)}(x) - b. \quad (25)$$

Lemma 9 (Pivot induction). For the true $K_t, h_{t+1}^{(j)}, b_t^{(j)}, \tilde{C}_{t-1}^{(j)} = C_{t-1}^{(j)}$.

Proof. Substitute (23) and $R_t = \text{E}_t(R_{t-1} + s_t)$ into (24):

$$W_{t-1} = R_{t-1} + b_t + 2^8 h_t.$$

Subtracting b_t gives (23) at depth $t-1$. □

The two guessed bytes come from consecutive state words: h is the high byte of s_{t+1} , already folded into the current pivot, whereas b is the low byte of s_t , removed after inversion. The high byte h_t that remains is carried automatically in the next pivot. This is why one invariant suffices at every depth.

At $t = 8$, $h_9 = 0$ is fixed and only b_8 is tried. At depths 7 through 2, the same pair and context-specific state bytes are tried across all contexts before descending to the next key.

8 Necessary conditions from carries

For a permutation F and quotient $\pi \in \{\text{hi}, \text{mid}\}$, define

$$\rho_\pi(F) = \max_u |\{\pi(F(x)) : \pi(x) = u\}|. \quad (26)$$

Thus $\rho_\pi(F)$ is the largest number of output quotient classes reached from any one input class.

Lemma 10 (One-translation lemma). *For every modular translation $\tau_s(x) = x + s$, $\rho_{\text{hi}}(\tau_s), \rho_{\text{mid}}(\tau_s) \leq 2$, and the same holds for τ_s^{-1} .*

Proof. Fixing the high byte leaves only the low byte variable. Addition of the low byte of s contributes either zero or one carry to the high byte. For the middle byte, write $x = 2^{12}a + 16m + d$. The low nibble contributes at most one carry into m ; the top nibble affects only discarded bits modulo 256. Negating s proves the inverse statement. $\square$

Theorem 11 (Bidirectional retention bound). *Let R_r be a cascade of r translated SEPAR stages and let $C_r = R_r + 2^8 h$. Then*

$$\rho_{\text{hi}}(C_r), \rho_{\text{mid}}(C_r), \rho_{\text{hi}}(C_r^{-1}), \rho_{\text{mid}}(C_r^{-1}) \leq \min(2^r, 256). \quad (27)$$

Proof. Each E_i and D_i maps a quotient fibre into exactly one quotient fibre by Corollary 3; it only relabels fibres. Each modular translation can split one fibre into at most two by Lemma 10. Induction through r stages gives 2^r , capped by 256 possible quotient values. Inverting the cascade reverses the order and replaces translations by inverse translations, so the same induction applies. The final multiple-of-256 translation is itself quotient-preserving for both listed quotients. $\square$

After a stage is peeled, retain a joint (h, b) candidate only when all four quantities in (27) meet the bound for $r = t - 1$. Theorem 11 guarantees that the true candidate survives. Wrong candidates may survive as well: these are necessary conditions, whereas the scores below only choose an order among survivors.

8.1 Efficient state-byte enumeration

Proposition 12 (Shift invariance of the implemented support statistics). *Fix a pivot table C , a candidate stage permutation E_K , and write*

$$W_h = D_K \circ \tau_{-2^8 h} \circ C, \quad 0 \leq h < 256.$$

For each $\pi \in \{\text{hi}, \text{mid}\}$, both $\rho_\pi(W_h)$ and $\rho_\pi(W_h^{-1})$ are independent of h , giving four invariant maxima in total. More strongly, the forward support size of every fixed input row is unchanged, and the multiset of inverse-row support sizes is unchanged.

Proof. For two guesses h, h' , direct cancellation gives

$$W_h = A \circ W_{h'}, \quad A = D_K \circ \tau_{2^8(h'-h)} \circ E_K.$$

Each factor of A preserves both quotient partitions, so A induces a permutation on their fibres. Postcomposition by A merely relabels the output fibres in every forward row and cannot change a row's support size. Furthermore $W_h^{-1} = W_{h'}^{-1} \circ A^{-1}$; here A^{-1} permutes the input rows, so the inverse-row support multiset is unchanged. Taking maxima proves the claim. $\square$

Proposition 13 (Exact all-byte sweep). *Let F be a permutation of $\mathbb{Z}/2^{16}\mathbb{Z}$ and $F_b = \tau_{-b} \circ F$, $0 \leq b < 256$. For every b , all four values*

$$\rho_{\text{hi}}(F_b), \quad \rho_{\text{mid}}(F_b), \quad \rho_{\text{hi}}(F_b^{-1}), \quad \rho_{\text{mid}}(F_b^{-1})$$

and the ordering statistic

$$U(F_b) = \sum_{u=0}^{255} |\{\text{hi}(F_b(x)) : \text{hi}(x) = u\}| \quad (28)$$

can be computed exactly in $O(2^{16})$ time and $O(2^{16})$ memory for all 256 values of b together.

Proof. Fix an input high-byte row and, for each output high byte q , record the minimum and maximum low residues occurring in that row. After subtracting b , output row q occurs precisely when an old- q residue is at least b , or an old- $(q+1)$ residue is less than b . These are two intervals of b -values. A difference-array sweep over the 256 thresholds therefore gives every forward row support, its maximum, and its contribution to U . For the middle-byte quotient the identical argument uses the low-nibble residue; the high nibble of b only relabels output fibres.

Build F^{-1} once. Since $F_b^{-1} = F^{-1} \circ \tau_b$, increasing b by one moves exactly one boundary column between adjacent high-byte input rows. Maintaining output-fibre multiplicities gives all inverse high-byte supports. The middle-byte calculation moves one boundary column for each of the 16 top-nibble values and depends only on $b \bmod 16$. Every update is an exact insertion or deletion. The threshold and sliding sweeps, including construction of F^{-1} , use $O(2^{16})$ exact updates and $O(2^{16})$ memory; the stored inverse dominates the memory bound. $\square$

Before b is fixed, any threshold on the four maxima treats all 256 values of h identically; the forward-row support sum U is invariant as well. The exact filter must therefore be evaluated jointly in (h, b) . The implementation forms $\tau_{-b} \circ W_h$, checks all four inequalities, and only then orders the survivors. This conclusion concerns these support statistics, not every possible labelled statistic of W_h .

Corollary 14 (Zero-low-byte orbit). *For fixed C, t, K , the 256 maps W_h from Proposition 12 have identical four retention maxima and identical values of U . Hence they are either all inadmissible or all admissible and tied. In particular, for the true pair when the true low byte is zero, all 256 high-byte guesses survive and tie with the true guess.*

Proof. For $b = 0$, the joint candidate after low-byte subtraction is W_h . Proposition 12 makes the four maxima independent of h . It also preserves the support size of each forward high-byte row, so their sum U is independent of h . For the true pair and low byte, Theorem 11 makes the true high byte admissible, and therefore makes the whole orbit admissible. $\square$

9 Candidate ordering

At each depth the search must order three finite sets: autonomous nibble permutations, raw key pairs within a nibble fibre, and surviving state-byte guesses. The first two scores count adjacent transitions in a quotient of the observed codebook. The last prefers candidate tables with smaller high-byte support. None of these scores rejects a candidate. Candidates with the same score form a *score tier*; a tier budget always visits the whole tie at its boundary.

Let J be the current set of active recovery contexts. For a pivot table C , a quotient π , and the fixed difference set Δ , form the edge matrix

$$M_{\pi, C}[u, v] = \#\{(x, d) : d \in \Delta, \pi(C(x)) = u, \pi(C(x+d)) = v\}. \quad (29)$$

For a lane tuple ν , lane offset λ , and raw pair K , let

$$\text{sc}_4(C; \nu, \lambda) = \sum_{z=0}^{15} M_{\text{lane}, C}[q_\nu(z) + \lambda, q_\nu(z+1) + \lambda], \quad (30)$$

$$Q_K(z) = \text{hi}(\mathbf{E}_K(2^8 z)),$$

$$\text{sc}_8(C; K, h) = \sum_{z=0}^{255} M_{\text{hi}, C}[Q_K(z) + h, Q_K(z+1) + h]. \quad (31)$$

Here $2^8 z$ is simply a representative of the high-byte fibre z ; Corollary 3 makes the choice of its low byte immaterial. Indices are reduced modulo 16 or 256. The implemented lane and pair scores are exactly

$$\text{Sc}_4(\nu) = \sum_{j \in J} \max_{0 \leq \lambda < 16} \text{sc}_4(C_t^{(j)}; \nu, \lambda), \quad (32)$$

$$\text{Sc}_8(K) = \sum_{j \in J} \max_{0 \leq h < 256} \text{sc}_8(C_t^{(j)}; K, h). \quad (33)$$

Thus every lane score examines all 16 translations and every pair score examines all 256 high-byte translations. There is no translation budget.

For a pair K , define

$$H_{K,j} = \arg \max_{0 \leq h < 256} \text{sc}_8(C_t^{(j)}; K, h).$$

Let

$$M_{16,C}[u, v] = \#\{(x, d) : d \in \Delta, C(x) = u, C(x + d) = v\},$$

and, with $x + 1$ reduced modulo 2^{16} , define

$$\text{sc}_{16}(C; K, h) = \sum_{x \in \mathbb{Z}/2^{16}\mathbb{Z}} M_{16,C}[\mathbf{E}_K(x) + 2^8 h, \mathbf{E}_K(x + 1) + 2^8 h]. \quad (34)$$

The implementation computes this sparse full-word score only for $h \in H_{K,j}$, and uses the aggregate

$$\text{Sc}_{16}(K) = \sum_{j \in J} \max_{h \in H_{K,j}} \text{sc}_{16}(C_t^{(j)}; K, h). \quad (35)$$

This Sc_{16} value orders candidates *inside one complete tie of Sc_8* . It neither defines a fourth budget nor removes a candidate: every member of a visited Sc_8 -tier is tried, regardless of its Sc_{16} order.

For a fixed pair K and context j , define

$$T_{t,K}^{(j)}(h, b) = \tau_{-b} \circ D_{t,K} \circ \tau_{-2^8 h} \circ C_t^{(j)} \quad (36)$$

and its exact admissible set

$$\mathcal{A}_{t,K}^{(j)} = \{(h, b) : \rho_\pi(T_{t,K}^{(j)}(h, b)), \rho_\pi((T_{t,K}^{(j)}(h, b))^{-1}) \leq \min(2^{t-1}, 256) \text{ for } \pi \in \{\text{hi}, \text{mid}\}\}. \quad (37)$$

Here $h = h_{t+1}^{(j)}$ and $b = b_t^{(j)}$. The packed value $2^8 h + b$ only encodes a candidate pair of bytes; it is not one SEPAR state word. Members of $\mathcal{A}$ are ordered by increasing $U(T_{t,K}^{(j)}(h, b))$ from (28). At the root, $h_9 = 0$ is fixed, so the analogous list contains only admissible b_8 .

The inward search has three tier boundaries; the outer phase has a separate limit on last-stage lanes:

1. visit score tiers of the 2^{16} lanes under Sc_4 ;
2. expand each visited lane into its exact raw-pair fibre and visit score tiers under Sc_8 , with sc_{16} used only as a within-tier order;
3. for each pair and active context, visit score tiers of its exact (h, b) survivors under U , or of b_8 at the root.

The three budgets count distinct score values, rather than individual candidates. A zero budget denotes all tiers. With all three budgets zero, the candidate set is independent of every score and tie order.

9.1 Context retirement in bounded search

Corollary 14 explains why expanding every tied state candidate can dominate a bounded run. The implementation therefore applies one deterministic rule when the state-tier budget is nonzero. When the lowest- U tier for the current context contains more than one candidate and at least two contexts remain active, that context is retired for the remainder of the current branch. The root applies the same rule to its lowest-support b_8 tier. A retired context contributes neither later ranking scores nor reconstructed state constraints; its status is restored on backtracking. At least one context always remains active. All input to this decision is computed from public codebooks and the current public candidate.

Lemma 15 (Exact-filter monotonicity under retirement). *Let $J' \subseteq J$. On the branch containing the true key and true pivots, restricting the componentwise exact tests from J to J' cannot reject the true key path on any retained context. In particular, retirement cannot invalidate pivot induction or the carry and stage-one tests that remain.*

Proof. For every $j \in J$, Lemma 9 supplies the true next pivot and Theorem 11 places its true (h, b) in $\mathcal{A}_{t,K}^{(j)}$. The exact multi-context condition is a conjunction, equivalently a Cartesian product of the per-context survivor sets. Deleting a component leaves the restriction of the true vector in the remaining product. The same argument applies at the root, and the true stage-one factor satisfies (39) separately in every retained context. $\square$

Lemma 15 is not a theorem about bounded ranking. After a retirement, the program rebuilds edge tables and recomputes Sc_4 and Sc_8 over the smaller active set. A true lane, pair, or state coordinate may then lie beyond a nonzero tier budget. Moreover, the rule can retire a context without knowing the rank of its true coordinate. The success probability and runtime of this bounded, dynamically re-ranked algorithm must therefore be evaluated end to end.

In exhaustive search the state budget is zero, so retirement is disabled. Every context remains active, every score tier is visited, and changing a score or tie order changes only the enumeration order, not the finite set of leaves.

9.2 Stage-one factor

At the final pivot,

$$C_1^{(j)}(x) = \mathbf{E}_{1,K_1}(x + s_1^{(j)}) + 2^8 h_2^{(j)}. \quad (38)$$

For every candidate (K, h) , test whether

$$\mathbf{D}_{1,K}(C_1^{(j)}(x) - 2^8 h) - x \quad (39)$$

is constant for all 2^{16} inputs. If so, the constant is the candidate $s_1^{(j)}$. The implementation enumerates all 256 values of h in every context, rather than retaining the first factor. For the true pair and true $h_2^{(j)}$, (39) is identically $s_1^{(j)}$.

10 Completeness

Theorem 16 (Eventual completeness and order independence). *For every key, deterministic IV family, and transcript produced by the implemented cipher, consider the traversal that enumerates all last-stage lanes and their raw-key fibres, uses zero inward tier budgets, never retires a context, and continues after accepted leaves until every finite loop is exhausted. It visits the original numeric key and its initialized states after finitely many candidates, and that leaf reproduces the transcript. Replacing an ordering score or a tie order changes only the order of this traversal, not its coverage.*

Proof. The outer traversal visits all 2^{16} lane tuples and every raw pair in each disjoint fibre, so Proposition 6 guarantees that it supplies the true K_8 . All three inward tier budgets are zero and retirement is disabled. For the true K_8 , Lemma 9 gives $\mathbf{D}_{8,K_8}(C_8^{(j)}) - b_8^{(j)} = C_7^{(j)}$, and Theorem 11 with $r = 7$ makes every true $b_8^{(j)}$ admissible. The root visits every admissible low byte with fixed $h_9^{(j)} = 0$, and hence reaches all true $C_7^{(j)}$ pivots. At each stage $t = 7, \dots, 2$, all lane tuples and every pair in their fibres are visited. For the true pair, Lemma 9 maps the true bytes $(h_{t+1}^{(j)}, b_t^{(j)})$ to the true next pivot, while Theorem 11 puts that byte pair in the exactly computed $\mathcal{A}_{t,K}^{(j)}$. A zero state-tier budget visits every member of every such set. Induction therefore reaches the true stage-one pivots in all contexts. Complete lane and pair traversal visits the true K_1 , all 256 values of every $h_2^{(j)}$ are tested, and (39) returns each true $s_1^{(j)}$.

Every loop ranges over a finite set. Scores are used only to sort or partition sets that the exhaustive search traverses in full, so altering their order does not alter coverage. The true key and states reproduce initialization, every cached inward codebook, and both held-out transcripts, hence the original-key leaf is accepted. $\square$

The proof concerns a traversal that continues after finding a candidate. The supplied program instead stops at the first key that reconstructs the active initializations and reproduces every collected codebook and both held-out 64-word transcripts. Such a key agrees with those finite transcripts. The inward leaf check does not replay the outer prefix transcript; that transcript orders candidates and supplies K_8 . The theorem guarantees that continued exhaustive enumeration reaches the original key; it does not assert that a finite transcript uniquely identifies its numerical representation.

This is an eventual enumeration theorem, not a polynomial-time or minute-scale bound. For c contexts, a deliberately coarse unfiltered leaf bound is

$$2^{32} 2^{8c} (2^{32+16c})^6 2^{32+8c} = 2^{256+112c}. \quad (40)$$

The factors count the outer pair, root low bytes, six intermediate pairs with two state bytes per context, and the stage-one pair and high bytes. The bound proves finiteness but is too loose to guide an implementation; exhaustive search over the 2^{256} master keys is a tighter unconditional fallback. The practical claim concerns bounded score tiers and is therefore statistical.

11 Data and computational complexity

The outer experiment contains exactly

$$7 \cdot 2^{16} = 458,752 \quad (41)$$

separately reset chosen-plaintext messages of two words each, hence 917,504 online word encryptions. It contributes $6 \cdot 2^{16} = 393,216$ directed nonzero-difference edges.

For each of c deterministic IVs, inward recovery resets before every input x and collects a 2^{16} -entry one-word codebook. It therefore uses $c2^{16}$ reset messages and the same number of word encryptions. Verification adds two held-out reset messages, each carrying a stateful 64-word transcript under a distinct IV. Every response is cached before search, so backtracking and failed leaves require no further queries. For $c = 8$, the stand-alone inward phase uses 524,290 messages containing 524,416 words. If the two phases are collected separately, their sum is 983,042 messages containing 1,441,920 words. Choose the fixed outer IV as the first member of the inward IV family. Each outer two-word response already contains the first ciphertext word $R_8(p)$, so retaining one response for every prefix supplies that inward codebook at no extra query cost. A cached implementation therefore uses

$$\begin{aligned} Q_{\text{msg}} &= 458,752 + (c - 1)2^{16} + 2, \\ Q_{\text{word}} &= 917,504 + (c - 1)2^{16} + 128. \end{aligned} \quad (42)$$

For $c = 8$, these are 917,506 messages containing 1,376,384 words. All responses are cached before the search, so neither count has a per-leaf factor.

A black-box implementation can collect the fixed outer experiment, the remaining codebooks, and the held-out transcripts once, then cache them for the whole search. The dominant offline costs are summarized below. Here N is the number of raw pairs in the selected outer fibres, B is the number given a full-word outer score, and $a = |J|$ is the number of active inward contexts.

Table 2: Dominant offline costs at each level of the search.

Operation	Work
Outer high-byte lift	$2^8 N$ stage evaluations
Outer full-word refinement	$2^{16} B$ stage evaluations
Inward lane ordering	$a2^{24}$ matrix additions
Inward pair ordering in a fibre of size N_ν	$2^8 N_\nu$ stage calls and $a2^{16} N_\nu$ additions
Joint state-byte sweep, per pair and context	2^{24} inverse-stage calls and $O(2^{24})$ updates
Root low-byte sweep, per context	$O(2^{16})$
Stage-one factor test, per (K, h)	at most 2^{16} inverse-stage calls

One lane contains at most 331,776 raw pairs. Enumerating every outer lane gives coarse bounds 2^{40} and 2^{48} for the high-byte and full-word passes. The joint state-byte sweep dominates a typical bounded inward node. Its 256 high-byte guesses are divided into contiguous ranges processed by native threads and merged in a deterministic order. These are costs per visited candidate or tier; the number of branches reached by bounded search is an empirical quantity.

12 Experimental evaluation

The proof of concept consists of two portable threaded C programs, one for the last-stage prefix experiment and one for inward recovery. Their shared cipher core is a literal implementation of the reference operations. Self-tests check the published stream, block inversion, the lane formula and exhaustive lane injectivity, the optimized carry computation against a direct reference, pivot induction, and the stage-one factor test. A bounded search that exhausts its configured tiers reports **INCONCLUSIVE**, rather than claiming failure of the full attack.

We evaluated 100 independently generated 256-bit keys, fixed before the runs. The inward experiment used eight deterministic IV contexts, seed one, a 600-second timeout, and tier budgets

$$(\text{lane}, \text{pair}, \text{state}) = (1, 1, 1).$$

It was conditioned on the correct externally supplied K_8 , so it measures the inward phase rather than time spent trying earlier last-stage candidates. A run counted as successful only if it passed transcript verification and returned the sampled numeric key. Every timeout, inconclusive run, or other output counted as a failure. The outer experiment was then evaluated on the same keys and fixed IV, retaining four lane ranks.

The inward phase recovered 95 keys, with three inconclusive runs and two timeouts. The outer phase placed the correct K_8 first for all 100 keys. Consequently the matched event “outer first and inward success” also occurred 95 times. Table 3 gives exact Clopper–Pearson intervals [1]. The reported 95th percentile is the nearest-rank order statistic.

Table 3: Bounded-search results on the same 100 keys. Inward recovery is conditional on the correct external K_8 ; its timing columns describe the 95 successful runs.

Experiment	Exact recovery	Exact 95% interval	Median	$\hat{q}_{.95}$
Inward, given K_8	95/100	[0.8872, 0.9836]	181.1 s	284.1 s
Outer candidate first	100/100	[0.9638, 1]	—	—
Matched joint event	95/100	[0.8872, 0.9836]	—	—

The exact one-sided 95% lower confidence bound for inward and matched success is 0.8977; the corresponding upper bound on failure is 0.1023. Successful inward runs ranged from 73.5 to 326.9 seconds, with mean 183.4 seconds. Across all trials, including the two timeouts at their observed duration, the median was 181.0 seconds and the nearest-rank 95th percentile was 285.7 seconds.

The machine was an AMD Ryzen 9 8945HS with 16 logical CPUs, running Linux 6.18.33.2 under WSL2, GCC 11.4.0, and glibc 2.35. The evaluator ran two processes concurrently with eight native threads per process; the reported times therefore include their contention. They measure conditional inward recovery and verification, not data acquisition or end-to-end latency.

Under the model that the sampled keys are independent uniform draws and the resulting per-key strict-success indicators are Bernoulli trials, failure probability p_f with f failures among n keys has the exact one-sided Clopper–Pearson bound, for $f < n$,

$$p_f \leq \text{Beta}_{1-\alpha}^{-1}(f+1, n-f) \quad (43)$$

at confidence $1 - \alpha$ [1]. The success lower bound is its complement. Only one indicator per key enters this calculation; stage, context, and retirement ranks within a run are correlated and can be censored by early termination.

For all five missed keys, the true inward S-box lane lay outside the one-tier budget. It occurred in tier 3 at stage 6 for trial 15, and in tiers 8, 9, 4, and 2 at stage 7 for trials 50, 54, 89, and 91. Whenever the corresponding branch was reached, every observed true byte, raw pair, and joint state-byte coordinate ranked first. These within-run ranks are correlated and censored by early stopping; they explain the observed failures but are not independent probability estimates.

The outer result does not establish universal rank one; the certificate in Section 6 disproves such a statement. The matched 95/100 result is the Boolean intersection of the two outcomes on identical keys, not a product of estimated rates. It also does not measure the time required to run inward recovery on false K_8 candidates that precede the true pair.

13 Discussion and conclusion

SEPAR’s sequential assignments collapse the intended diffusion layer and make every keyed stage triangular. Four proper coordinate partitions are preserved exactly, one nibble evolves autonomously, and only a tiny structured subset of all 16-bit permutations can occur. The OR-based key derivation and the use of disjoint key pairs make these properties exploitable. Separately, initialization reaches only 128 bits of the nominal 144-bit state.

The resulting attack is a factorization of the cipher from the outside in. It enumerates last-stage key fibres, maintains one pivot while peeling the remaining stages, discards only candidates that fail a proved carry bound, and closes with an exact stage-one identity. An exhaustive traversal is complete and independent of its scores. The bounded version replaces exhaustive ordering by three small score-tier budgets and context retirement; its minute-scale behavior is therefore supported by the experiment, not by an unproved maximality assumption.

The result is limited to the repeated-reset, chosen-IV setting of Section 2. We do not give a useful universal $\text{sub-}2^{256}$ time bound, so the completeness theorem does not imply that every key is recovered in minutes. The exhaustive proof and the measured score ordering are therefore separate claims.

A redesign must replace the linear layer with one having a justified branch number and no closed coordinate partition, state simultaneous operations unambiguously, avoid the lossy OR derivation, and diffuse key material between stages. A 16-bit external permutation also makes complete codebooks cheap, so enlarging only the internal state is not sufficient. Statistical output tests cannot substitute for this structural analysis.

The accompanying artifact contains the two C programs, the reference cipher source, the complete 100-key result table, and representative execution logs.

A Attack traversal pseudocode

Here VISITTIERS visits every row in each selected score tier, including the whole boundary tie; budget zero selects all tiers. Lane and pair tiers are visited in decreasing score order, while state tiers are visited in increasing U order. STATELIST enumerates all 2^{16} joint byte pairs, discards only failures of the four proved bounds, and sorts the survivors by increasing U . ROOTLIST does the same for the 256 low bytes with $h_9 = 0$. Sets J are immutable in the pseudocode, so return from a call automatically restores a retired context.

```
Recover(t, C[1..c], J):
  LaneScore(nu) = sum(j in J) max(lambda=0..15) score4(C[j], nu, lambda)
  for lane tier LN in VisitTiers(all nu by decreasing LaneScore, lane_budget):
    for nu in every row of LN:
      PairScore(K) = sum(j in J) max(h=0..255) score8(C[j], K, h)
      for every K in ExactFibre(nu)
        for pair tier PK in VisitTiers(all K by decreasing PairScore, pair_budget):
          order every K in PK by decreasing score16; do not remove any K
          for K in that order:
            if t == 1:
              Factor(1, K, C, J)
            else:
              Advance(t, K, 1, C, J)
```

```
Advance(t, K, j, C, J):
  if j > c: Recover(t-1, C, J); return
  if j not in J: Advance(t, K, j+1, C, J); return
  A = StateList(t, K, C[j])
  F = the complete first score tier of A, if A is nonempty
  if A is nonempty and state_budget > 0 and |F| > 1 and |J| > 1:
    Advance(t, K, j+1, C, J - {j}); return
  for state tier T in VisitTiers(A, state_budget):
    for (h,b) in every row of T:
      Cnext[j] = D_t,K(C[j]-256*h)-b
      Advance(t, K, j+1, C with C[j]=Cnext[j], J)
```

```
Factor(j, K1, C, J):
  if j > c:
    if VerifyActiveInitialization() and VerifyEveryCachedCodebook()
      and VerifyBothHeldOutTranscripts():
      emit candidate; executable returns, conceptual complete search continues
  else if j not in J:
    Factor(j+1, K1, C, J)
  else:
    for h = 0..255:
      if D_1,K1(C[j](x)-256*h)-x is one constant s1 for every x:
        record s1 and h; Factor(j+1, K1, C, J)
```

```
RootAdvance(j, C, J):
  if j > c: Recover(7, C, J); return
  if j not in J: RootAdvance(j+1, C, J); return
  A = RootList(K8, C[j])
  F = the complete first score tier of A, if A is nonempty
  if A is nonempty and state_budget > 0 and |F| > 1 and |J| > 1:
    RootAdvance(j+1, C, J - {j}); return
  for state tier T in VisitTiers(A, state_budget):
    for b in every row of T:
      Cnext[j] = D_8,K8(C[j])-b
      RootAdvance(j+1, C with C[j]=Cnext[j], J)
```

```

Inward(K8):
  C8[j] = cached complete reset codebook for every j
  RootAdvance(1, C8, {1,...,c})

FullAttack:
  for nu in all 65536 outer lanes, ordered by the outer lane score:
    for K8 in every raw pair of ExactFibre(nu), ordered by byte/word scores:
      Inward(K8)

```

With all three budgets zero, retirement and score-tier pruning are absent, but the executable still stops at its first accepted leaf. For the coverage theorem, the accepted-leaf stop is disabled and the outer loop continues after each inward success. It therefore streams every raw pair from every lane fibre to a non-stopping inward traversal. With nonzero budgets, the same pseudocode is a dynamically re-ranked heuristic whose success rate must be measured.

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